Ans: A

Q4

Here the vertical forces are gravitational and magnetic forces. The horizontal forces are the electric force and air resistance.

The electrically charged oil drop is moving horizontally at constant velocity. The oil drop is in dynamic equilibrium and the resultant force is zero in all directions.

Vertically, the gravitational force is acting downwards. Therefore, the magnetic force must be acting upwards to ensure equilibrium.

The electric force cannot act vertically in this case because the electric field is horizontal, and air resistance should be acting in the direction opposite to the motion, which is horizontal in this case.

Ans: D

Q5